Net interest income excluding CIB's Markets businesses

In addition to reviewing net interest income on a managed basis, management also reviews net interest income excluding net interest income arising from CIB's Markets businesses to assess the performance of the Firm's lending, investing (including asset-liability management) and deposit-raising activities. This net interest income is referred to as non-markets related net interest income. CIB's Markets businesses are Fixed Income Markets and Equity Markets. Management believes that disclosure of non-markets related net interest income provides investors and analysts with another measure by which to analyze the non-markets-related business trends of the Firm and provides a comparable measure to other financial institutions that are primarily focused on lending, investing and deposit-raising activities.

The data presented below are non-GAAP financial measures due to the exclusion of markets related net interest income arising from CIB.

Year ended December 31, (in millions, except rates)	**2017**	2016	2015
Net interest income – managed basis[a][b]	**$ 51,410**	$ 47,292	$ 44,620
Less: CIB Markets net interest income[c]	**4,630**	6,334	5,298
Net interest income excluding CIB Markets[a]	**$ 46,780**	$ 40,958	$ 39,322
Average interest-earning assets	**$2,180,592**	$2,101,604	$ 2,088,242
Less: Average CIB Markets interest-earning assets[c]	**540,835**	520,307	510,292
Average interest-earning assets excluding CIB Markets	**$1,639,757**	$1,581,297	$ 1,577,950
Net interest yield on average interest-earning assets – managed basis	2.36%	2.25%	2.14%
Net interest yield on average CIB Markets interest-earning assets[c]	0.86	1.22	1.04
Net interest yield on average interest-earning assets excluding CIB Markets	2.85%	2.59%	2.49%

(a) Interest includes the effect of related hedges. Taxable-equivalent amounts are used where applicable.
(b) For a reconciliation of net interest income on a reported and managed basis, see reconciliation from the Firm's reported U.S. GAAP results to managed basis on page 52.
(c) The amounts in this table differ from the prior-period presentation to align with CIB's Markets businesses. For further information on CIB's Markets businesses, see page 65.

Calculation of certain U.S. GAAP and non-GAAP financial measures

Certain U.S. GAAP and non-GAAP financial measures are calculated as follows:

Book value per share ("BVPS")
Common stockholders' equity at period-end / Common shares at period-end

Overhead ratio
Total noninterest expense / Total net revenue

Return on assets ("ROA")
Reported net income / Total average assets

Return on common equity ("ROE")
Net income* / Average common stockholders' equity

Return on tangible common equity ("ROTCE")
Net income* / Average tangible common equity

Tangible book value per share ("TBVPS")
Tangible common equity at period-end / Common shares at period-end

* Represents net income applicable to common equity